

Organic Chemistry 2

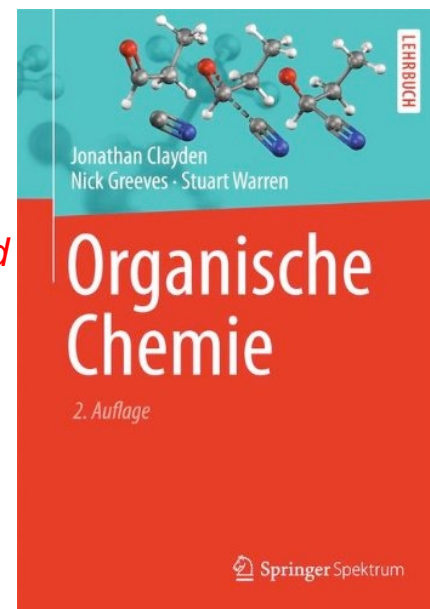
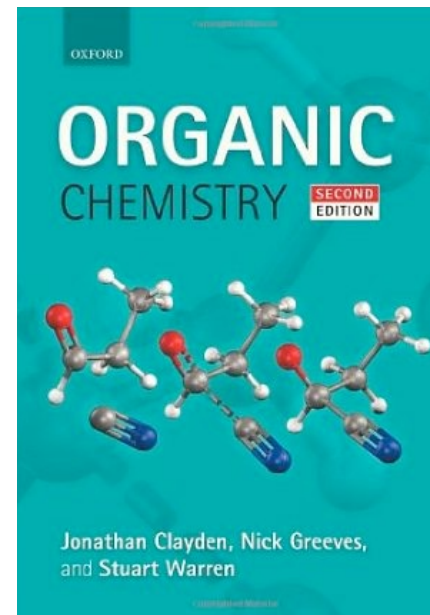
VO 347308

Winter 2026/27

mario.waser@jku.at

Textbooks

- Organic Chemistry
Paula Yurkanis Bruice - Pearson – Prentice Hall
- Organic Chemistry
L.G. Wade jr. - Pearson – Prentice Hall
- Organic Chemistry
K. Peter C. Vollhardt - Wiley-VCH
- Lehrbuch der Organischen Chemie
Hans Beyer, Wolfgang Walter - S. Hirzel Verlag
- Organic Chemistry
F.A. Carey, R.J. Sundberg - Wiley-VCH
- **Organic Chemistry**
Jonathan Clayden, Nick Greeves, Stuart Warren, Peter Wothers - Oxford Press (Deutsche Version bei Springer)
- Advanced Organic Chemistry
Jerry March - Wiley
- Strategy Applications of Named Reactions in Organic Synthesis
Laszlo Kürti, Barbara Czako - Elsevier



Content

- Einleitung – Struktur – Eigenschaften von org. Verbindungen
- Stabilität – Spannung – Resonanz
- Nomenklatur
- Stereochemie

- Fundamentals of Reactivity (Grundlagen der Reaktivität)
- Acid/Base Theory (Säure/Basen Theorie)
- Nucleophilic Substitution (SN) (Nukleophile Substitution)
- Elimination Reactions (Eliminierungsreaktionen)
- Electrophilic Addition to C=C (Elektrophile Addition)
- Reactivity of carbonyl and carboxylic acid derivatives (Carbonylchemie)
- Reactivity of Aromatic Compounds (Reaktivität von Arom. Verbindungen)
- Reactions of Radicals (Radikalreaktionen)
- Redox Chemistry (Redox Chemie)
- Rearrangements (Umlagerungen)
- Pericyclic Reactions (Pericyclische Reaktionen)
- Organometallic Chemistry (Metallorganische Reaktionen)

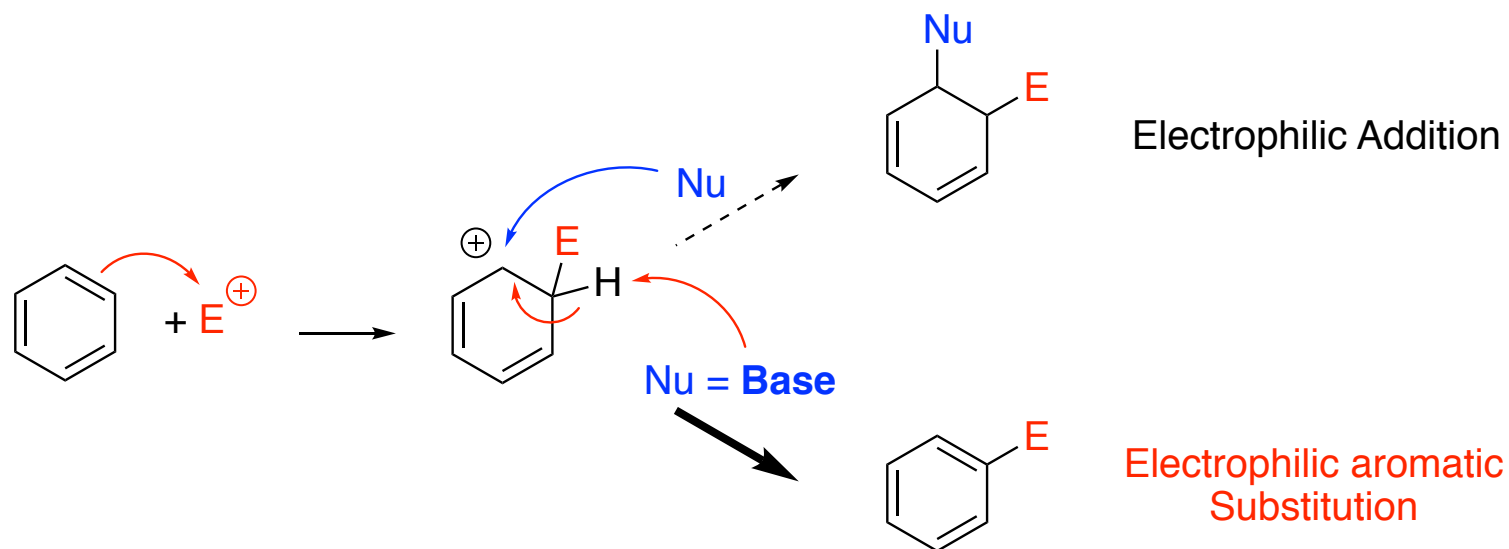
- Retrosynthesis (Syntheseplanung)

Reactivity of Aromatic Compounds

Electrophilic Aromatic Substitution

S_EAr: Most important reactivity mode of aromatic compounds!

A **Proton** is replaced by an **Electrophile** (the aromatic compound acts as a nucleophile)!

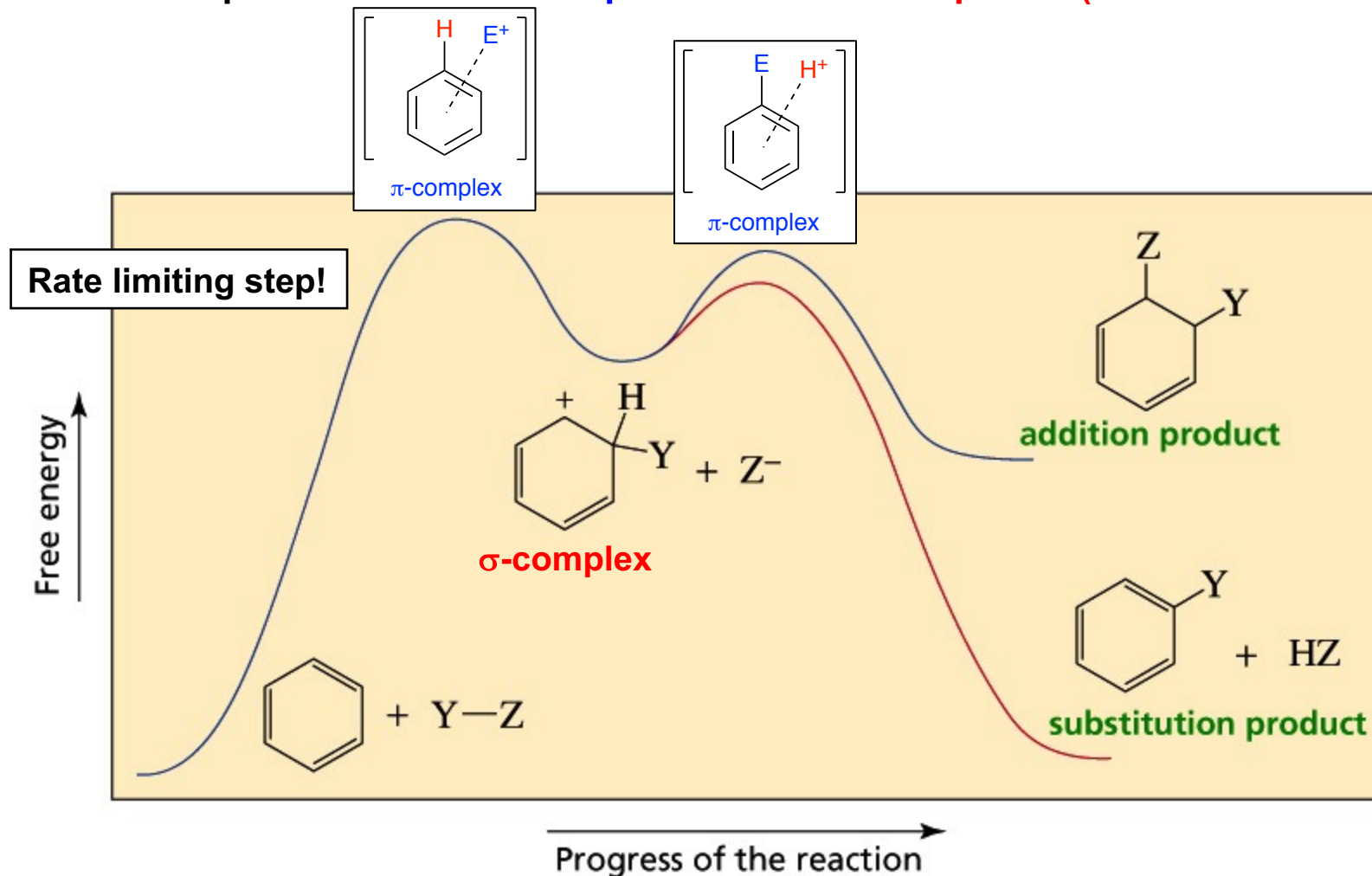


Electrophilic addition would lead to dearomatization!

Reactivity of Aromatic Compounds

Electrophilic Aromatic Substitution

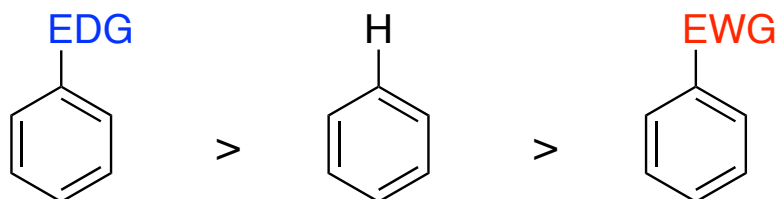
The reaction proceeds via 2 π -Complexes and 1 σ -Complexes (Wheland intermediate)!



Reactivity of Aromatic Compounds

Electrophilic Aromatic Substitution

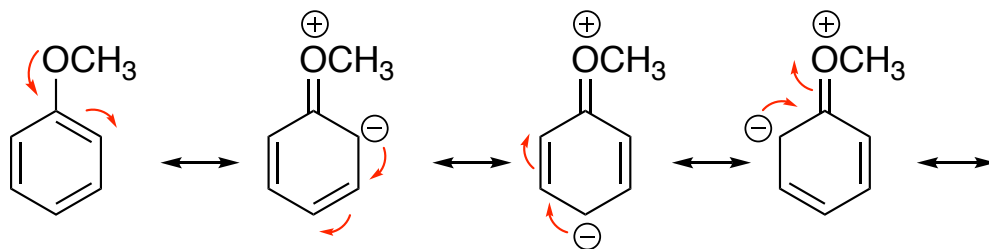
Influence of ring substituents on the **reaction rate**:



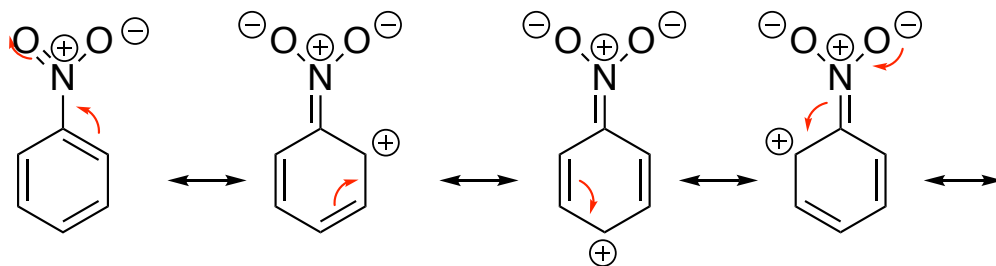
EDGs speed up (**activate**)!
EWGs slow down (**deactivate**)!

Comparison with H: **M-Effect** and **I-Effect**:

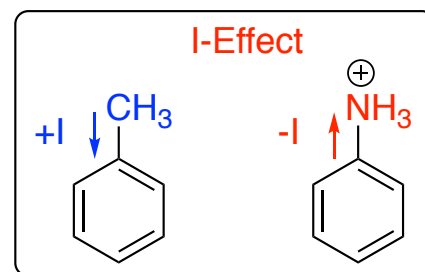
+M: Activation (e.g.: NR_2 , Alkyl, OR, ...)



-M: Deactivation (e.g.: NO_2 , COOH , CN , SO_3H , ...)



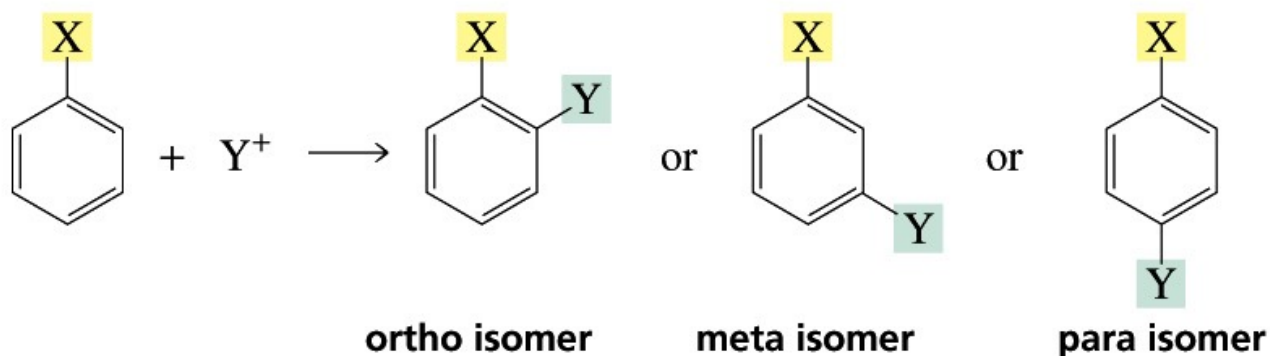
I-Effect



Reactivity of Aromatic Compounds

Electrophilic Aromatic Substitution

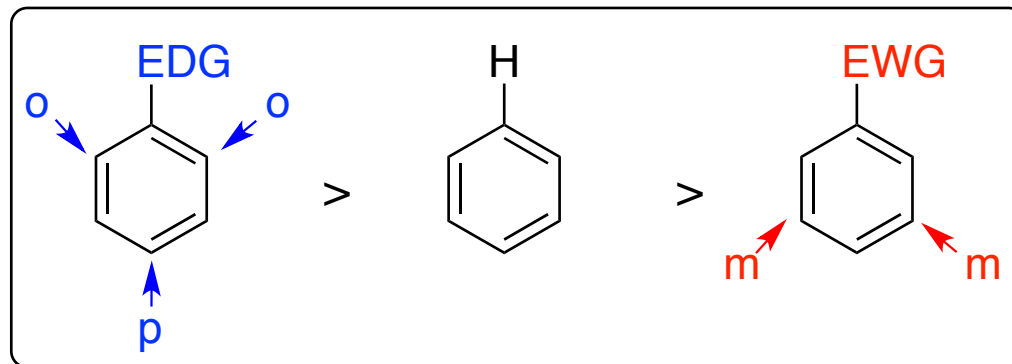
Influence of ring substituents on the **regioselectivity**:



EDGs give **ortho** and **para**!

EWGs give **meta**!

Overall substituent influence:

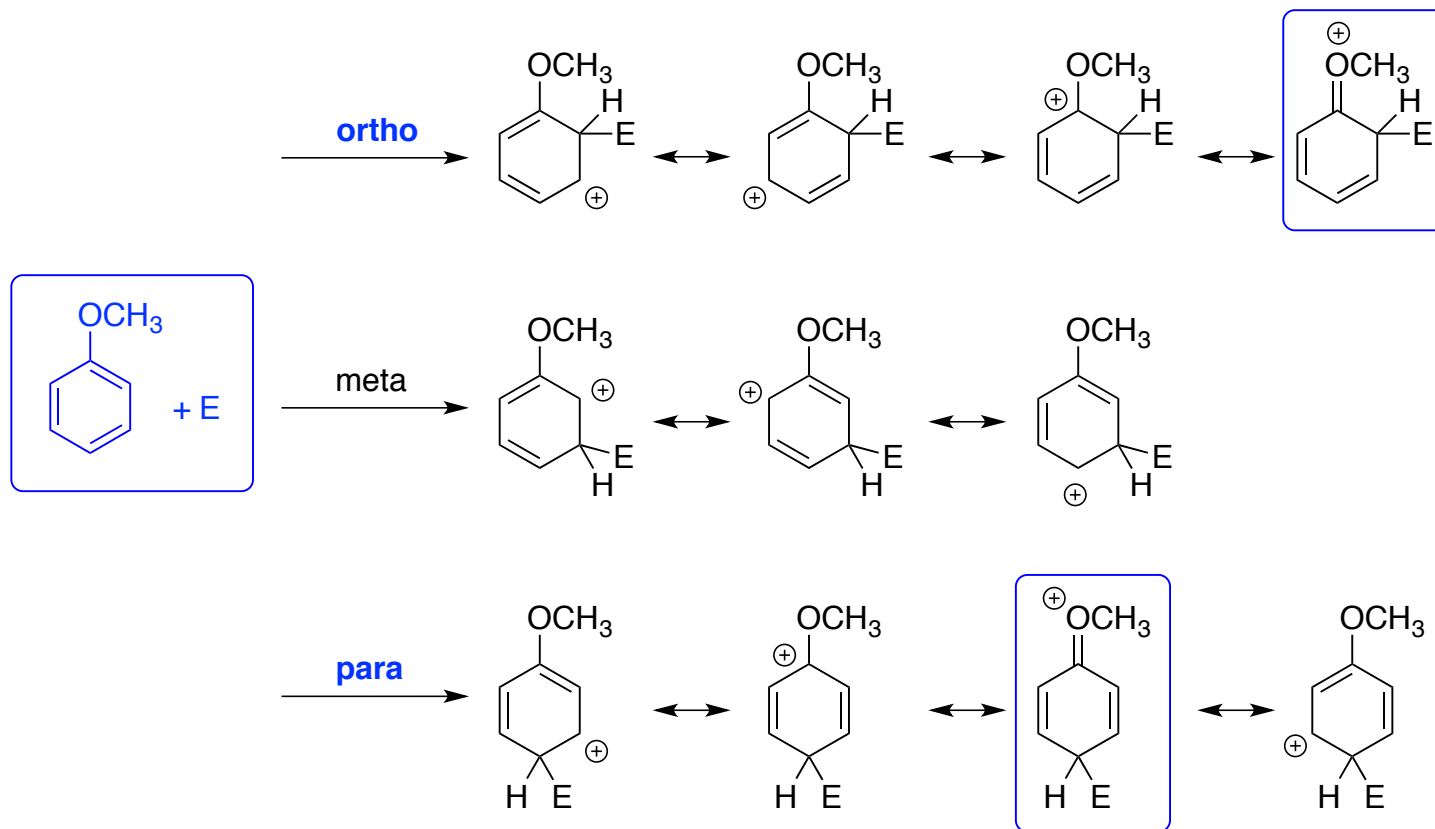


Reactivity of Aromatic Compounds

Electrophilic Aromatic Substitution

The observed regioselectivity can be rationalized by analyzing the **Wheland TS** (σ -Complexes):

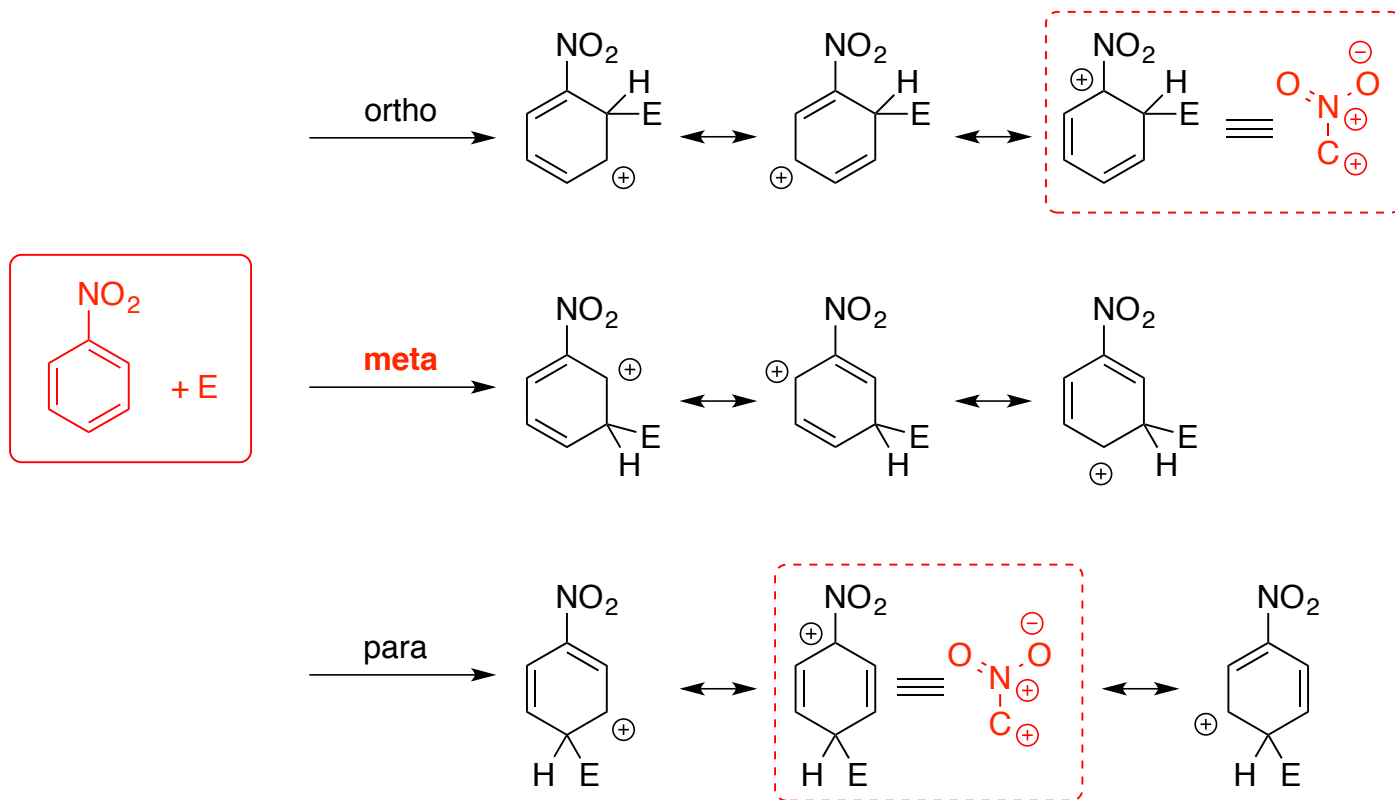
+M-Effect: o,p-directing!



Reactivity of Aromatic Compounds

Electrophilic Aromatic Substitution

-M and -I-Effect: 2 clearly less stable TS for ortho and para subst. → **meta directing!**



Halides are slightly deactivating (-I) but ortho/para-directing (+M)

Activating substituents	Most activating					
↓	—NH_2 —NHR —NR_2 —OH —OR	Strongly activating	Ortho/para directing			
	—NH—C(=O)—R —O—C(=O)—R			Moderately activating		
	—R —Ar —CH=CHR				Weakly activating	
	Standard of comparison →			—H		
	Deactivating substituents					
	↓	—F —Cl —Br —I		Weakly deactivating	Meta directing	
		—CH=O —C(=O)—R —C(=O)—COR				Moderately deactivating
		—COOH —CCl_3				
		$\text{—C}\equiv\text{N}$ $\text{—SO}_3\text{H}$ $\text{—NH}_2^+\text{R}$ —NR_3^+ —NO_2				
		—NH_3^+ —NHR_2^+				
		Most deactivating				

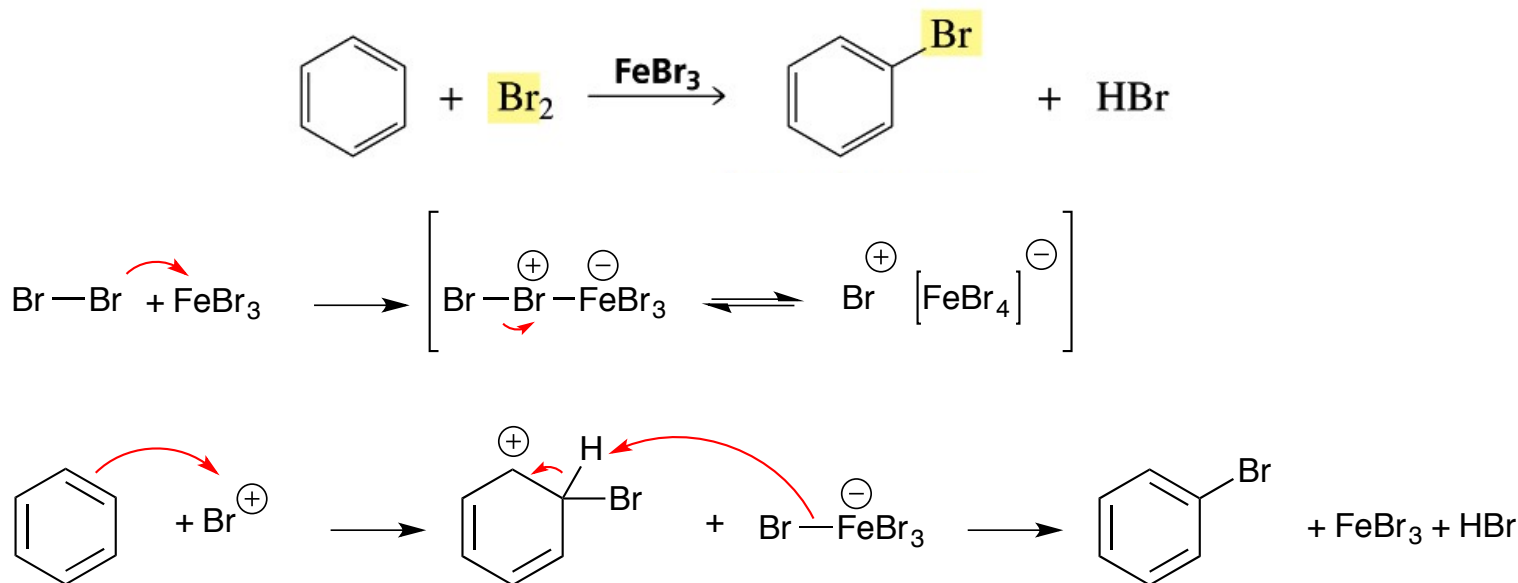
Reactivity of Aromatic Compounds

Electrophilic Aromatic Substitution

Common Electrophiles: X^+ , NO_2^+ , SO_3 , R^+ (most cases require catalysts to activate the electrophiles as most of the aromatic compounds are often too unreactive/less nucleophilic!)

Halogenation: **Lewis Acid** catalysis to activate the electrophile necessary!

Usually FeX_3 or AlX_3 – work well for **Cl/Br** but not suited for **F** (too reactive – other methods necessary) and **I** ($FeI_3 + I_2$ not reactive enough – alternative: $I_2 + AgNO_3$)
(only strongly activated aromatics react without catalyst)

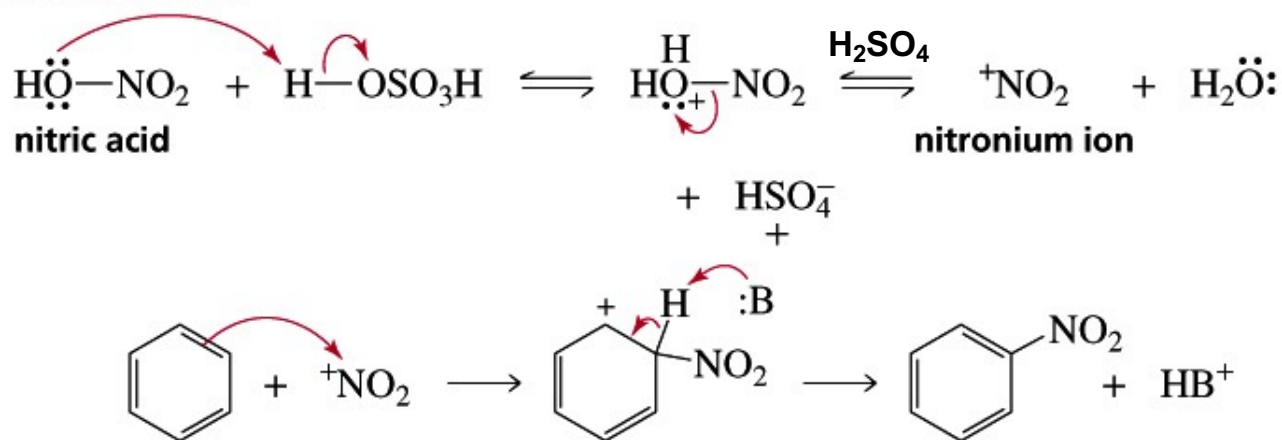
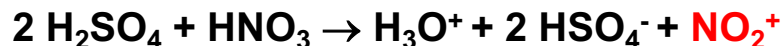


Formation of product mixtures and multiple substitutions possible!

Reactivity of Aromatic Compounds

Electrophilic Aromatic Substitution

Nitration: **Nitronium ion** as the electrophilic species – $\text{HNO}_3 + \text{H}_2\text{SO}_4$ (**Nitriersäure**)



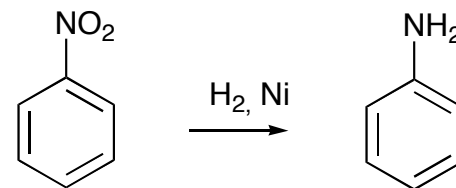
The nitration strength can be controlled by the amount of **NO₂⁺** in solution:

H_2SO_4 (95%) : HNO_3 = 2:1 → full NO_2^+ formation (very strong cond.)

HNO_3 (conc) → low concentration of NO_2^+

AcOH : HNO_3 → very little NO_2^+ only (mild cond.)

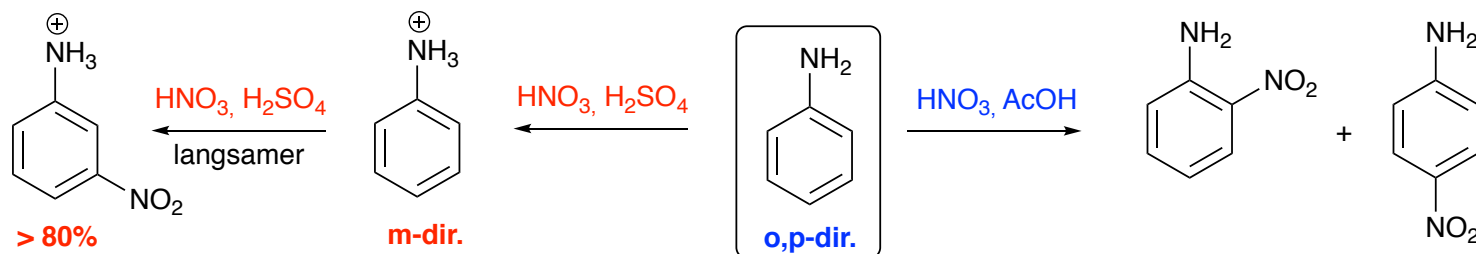
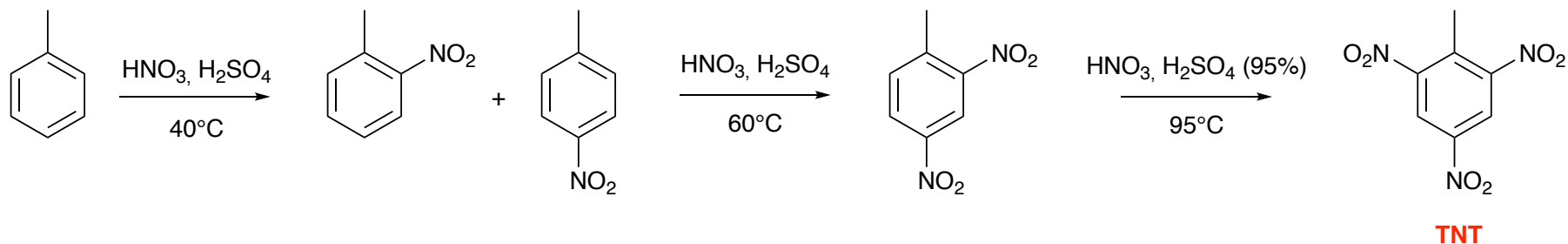
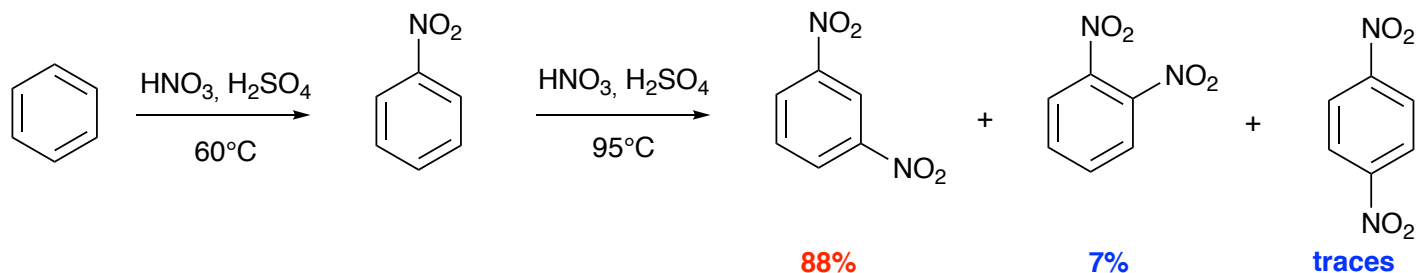
Nitro-Groups (i.e. aromatic) are readily hydrogenated.



Reactivity of Aromatic Compounds

Electrophilic Aromatic Substitution

Nitration: Different conditions allow for a control of the reaction outcome.



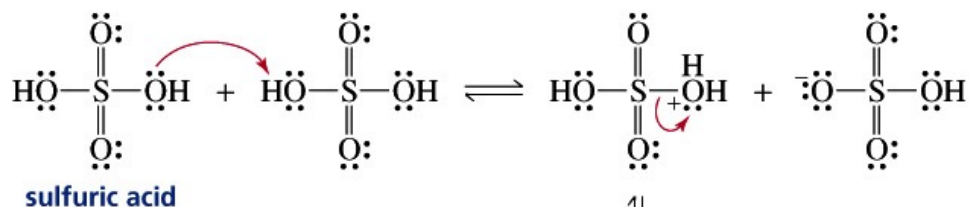
+ small quant. of o,p due to direct (fast) nitration of free-base aniline

Reactivity of Aromatic Compounds

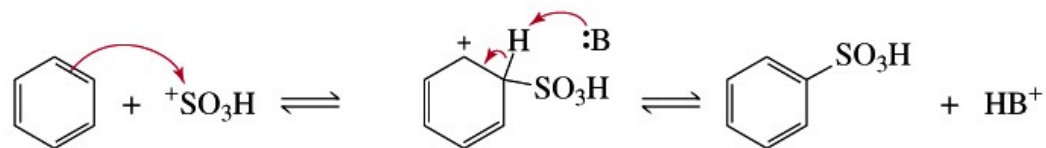
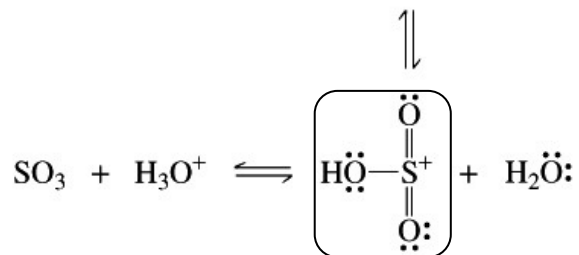
Electrophilic Aromatic Substitution

Sulfonation: Reversible Reaction:

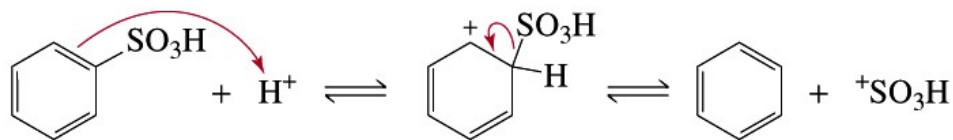
conc. H_2SO_4 gives sulfonation
40-60% H_2SO_4 leads to desulfonation



Conc. H_2SO_4 (Oleum):



Diluted H_2SO_4 :



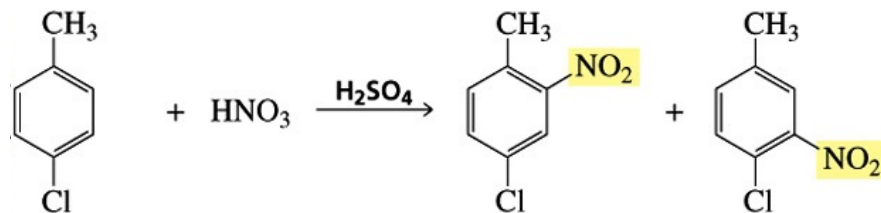
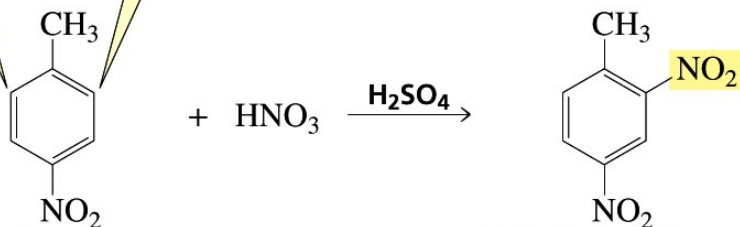
ipso-Substitution (Subst. of a functional group at the ring)

Reactivity of Aromatic Compounds

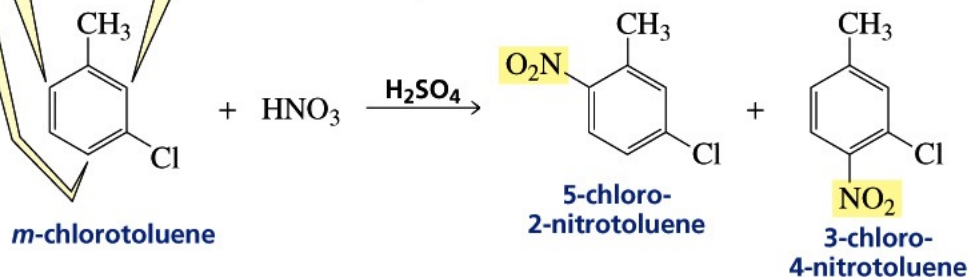
Electrophilic Aromatic Substitution

Examples for the regiocontrol in multiple-substituted substrates:

the methyl and nitro substituents direct the incoming substituent to the indicated positions



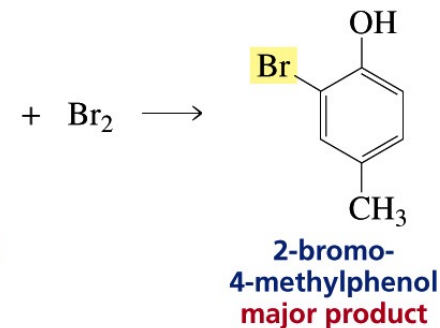
the methyl and chloro substituents direct the incoming substituent to the indicated positions



OH directs here

CH₃ directs here

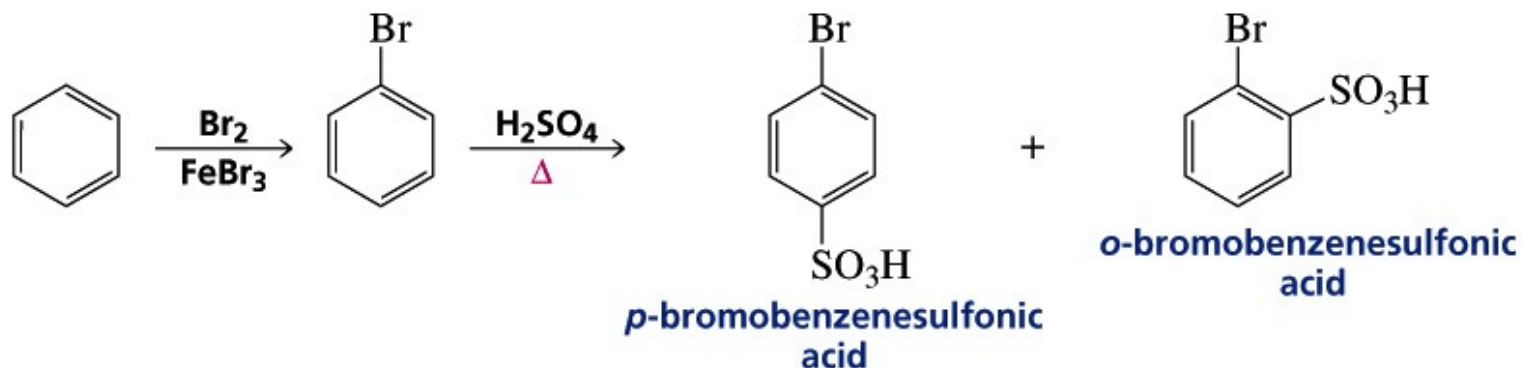
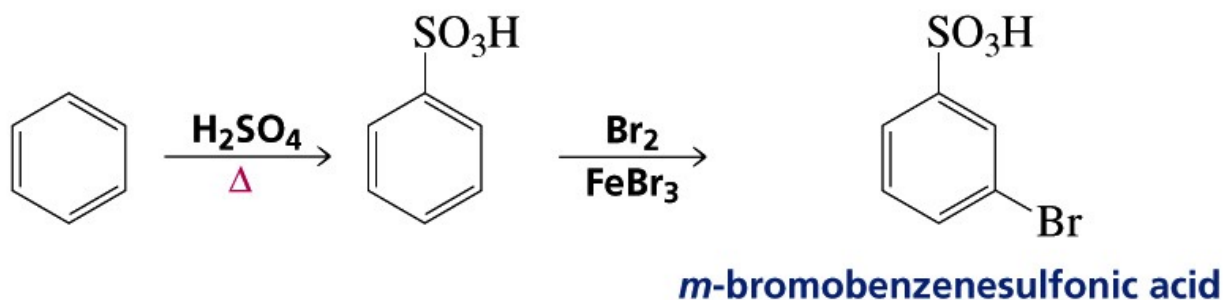
p-methylphenol



Reactivity of Aromatic Compounds

Electrophilic Aromatic Substitution

Regioselectivity in dependence on the order of addition steps:

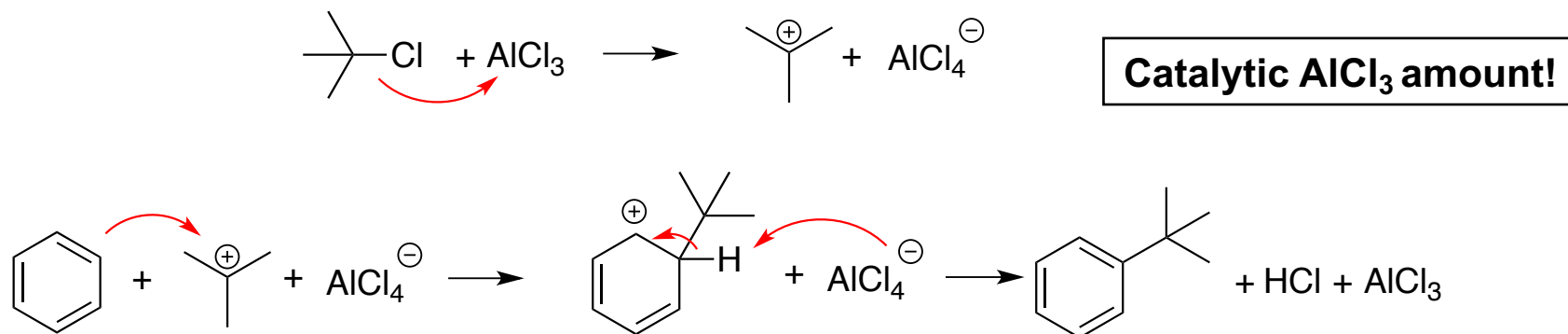


Reactivity of Aromatic Compounds

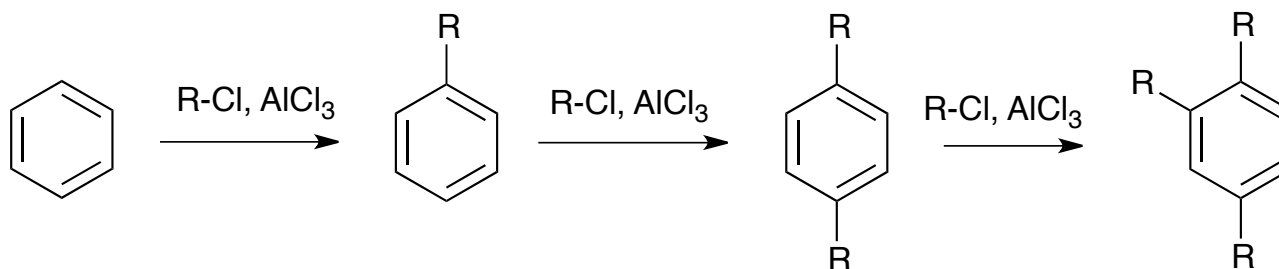
Electrophilic Aromatic Substitution

C-C Bond Formation:

Friedel Crafts Alkylation: Alkylhalides as electrophiles with neutral or activated aromatics in the presence of Lewis acids (i.e. AlCl_3)!



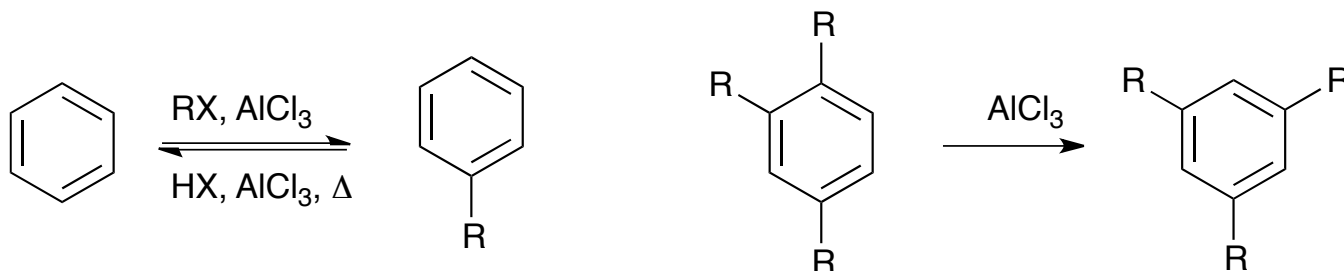
Friedel Crafts alkylations are often hard to control and tend to undergo multiple alkylations as the mono-alkylated product is usually more activated than the starting compound.



Reactivity of Aromatic Compounds

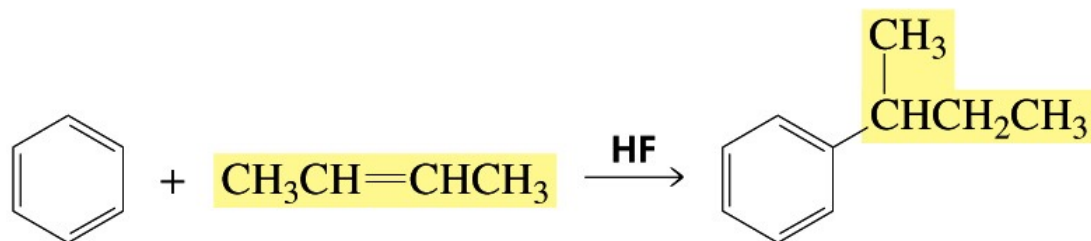
Electrophilic Aromatic Substitution

Friedel Crafts alkylations are in principle reversible and can lead to the rearrangement/migration of alkyl groups under prolonged reaction conditions!



Friedel Crafts reactions require neutral or activated substrates (strongly deactivated do not react at all!)

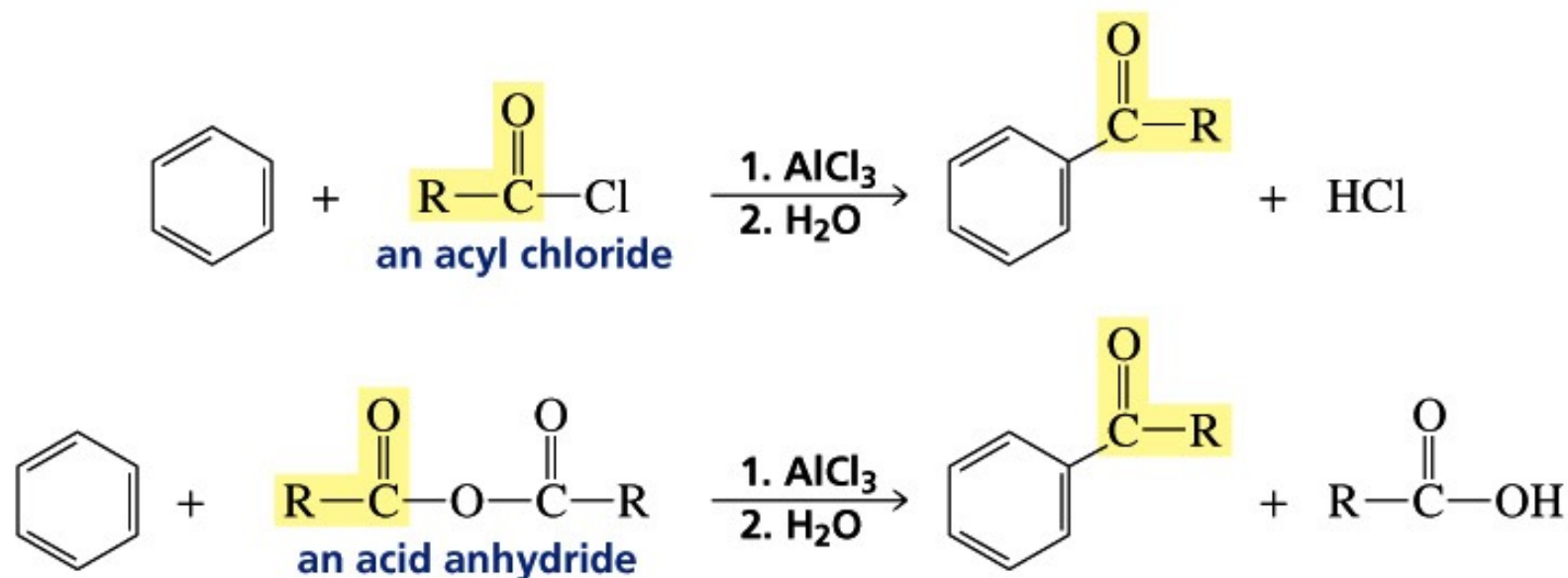
Acid-catalyzed alkylation using alkenes or alcohols also possible (via carbocation):



Reactivity of Aromatic Compounds

Electrophilic Aromatic Substitution

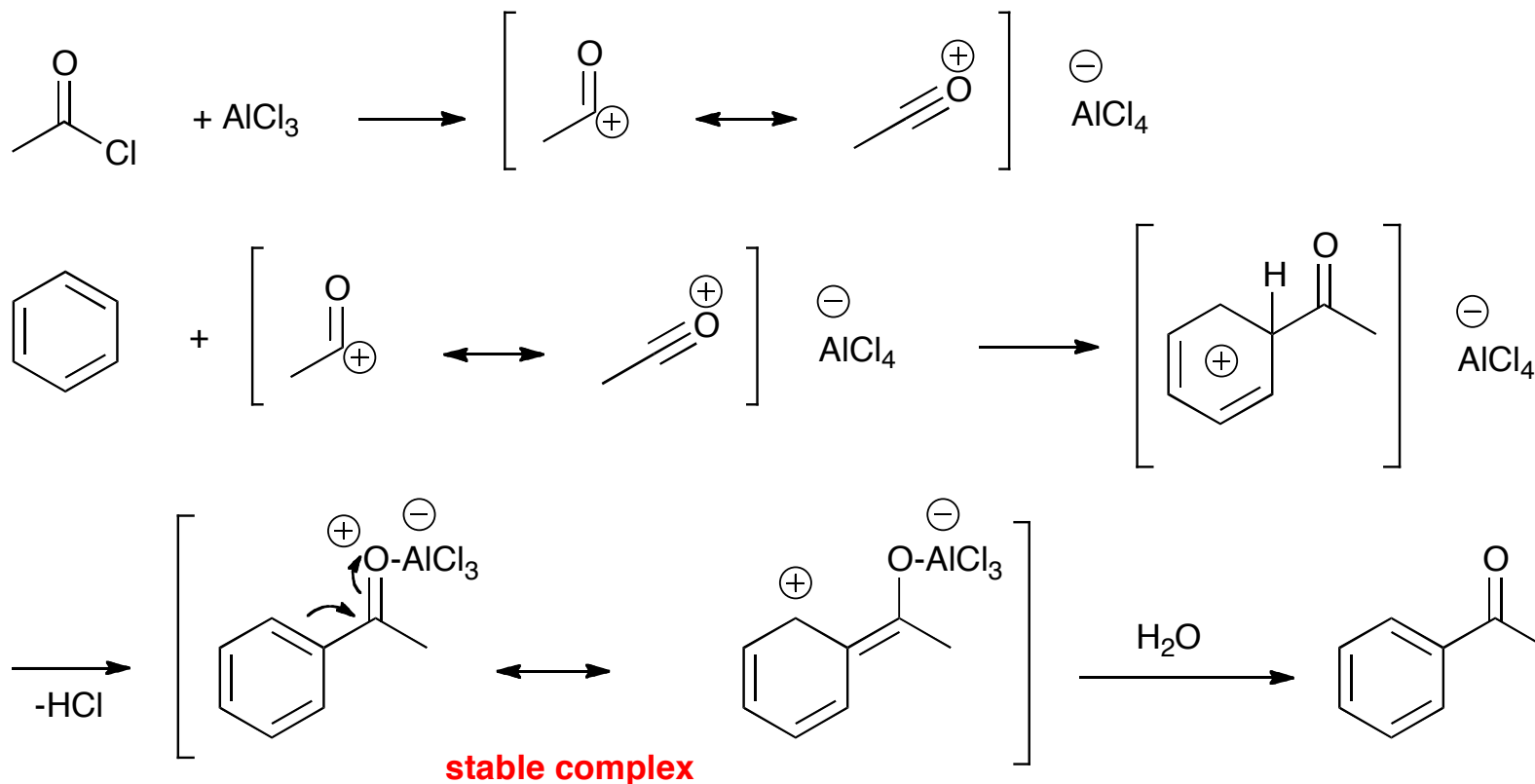
Friedel Crafts Acylation: Acylhalides or Anhydrides in the presence of a Lewis acid!



**Synthetically more useful and better to control than the alkylation
(Selective Monoacylation)!**

Reactivity of Aromatic Compounds

Electrophilic Aromatic Substitution

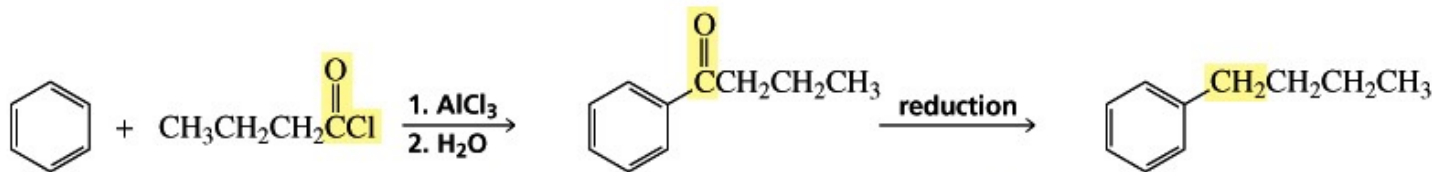


Lewis acid must be used in an equimolar amount (stable intermediate Al-complex)!

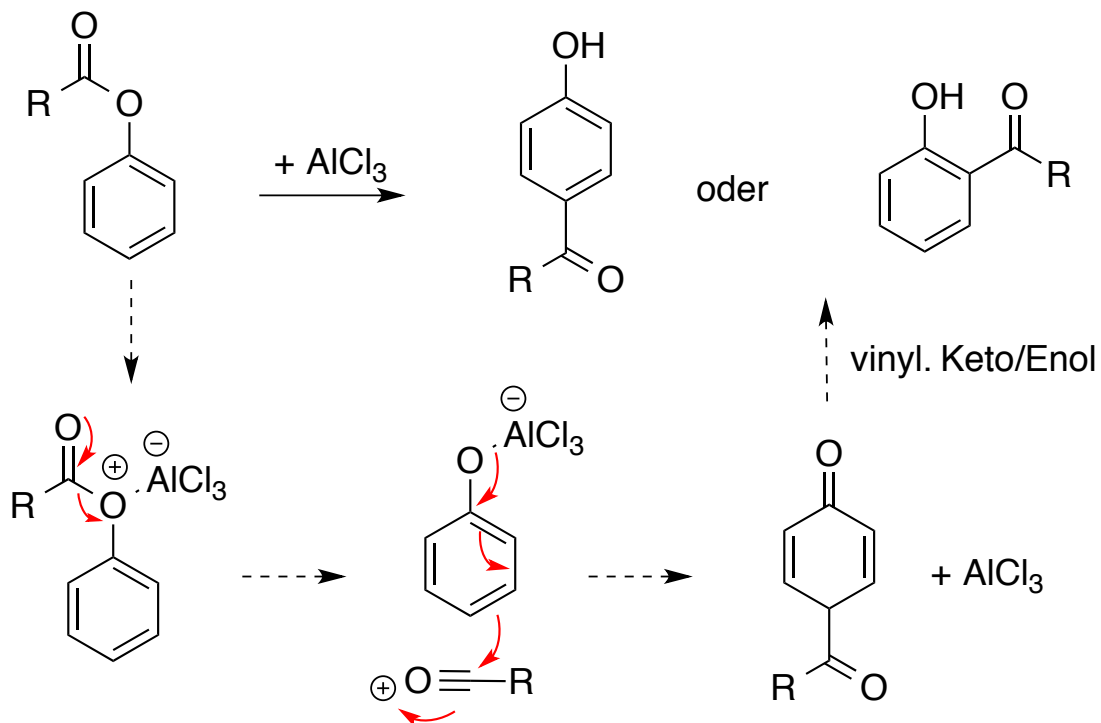
Reactivity of Aromatic Compounds

Electrophilic Aromatic Substitution

F.C. Acylation allows for the synthesis of products that are difficult by F. C. alkylations!



Fries Rearrangement:

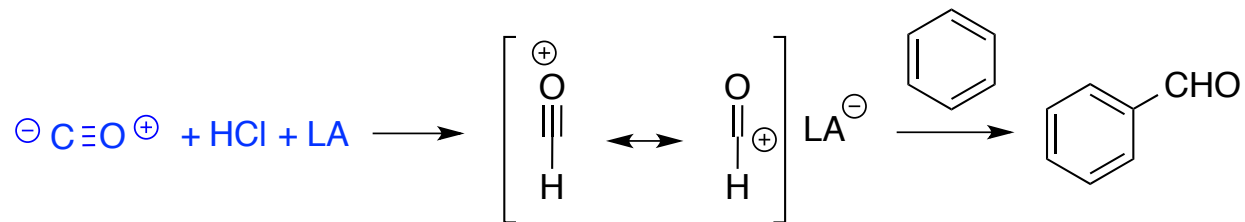


Reactivity of Aromatic Compounds

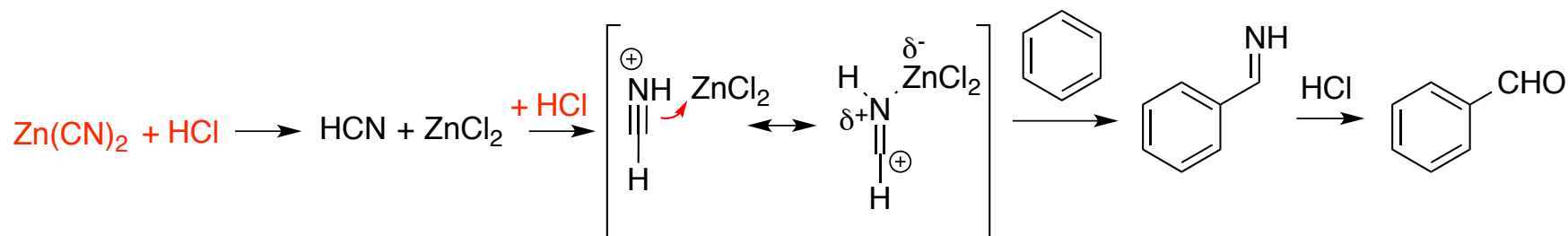
Electrophilic Aromatic Substitution

Formylation: Formylchloride and Formic acid anhydride are instable and not available and the required formyl electrophile HCO^+ has to be prepared in situ from other sources (mostly reactions with activated aryl substrates only!)

Gattermann-Koch Reaction: CO as formyl source!



Gattermann-Adams Reaction: $\text{Zn}(\text{CN})_2$ as formyl source (via HCN)!

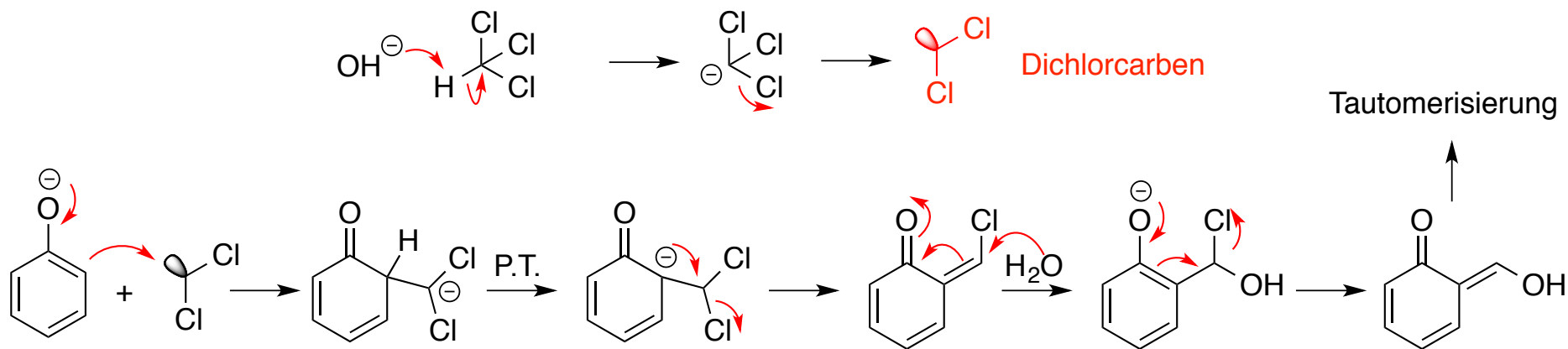
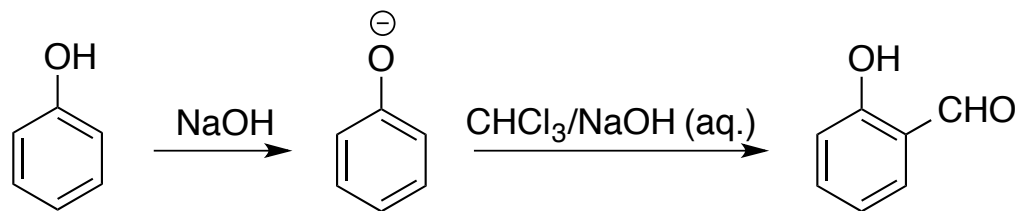


Houben-Hoesch: Analogous use of nitriles (RCN) for acylations!

Reactivity of Aromatic Compounds

Electrophilic Aromatic Substitution

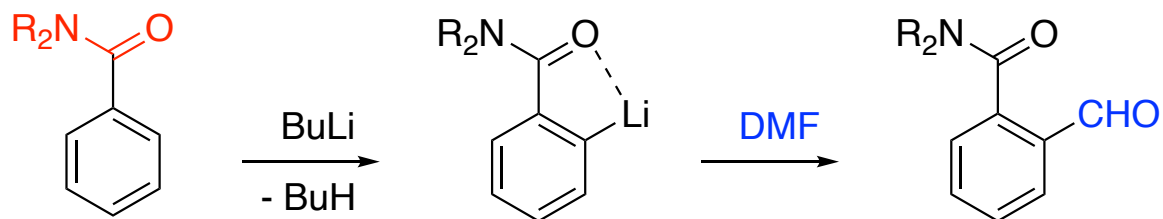
Reimer-Tiemann: Formylation of Phenols under **basic Cond.**!



Reactivity of Aromatic Compounds

Electrophilic Aromatic Substitution

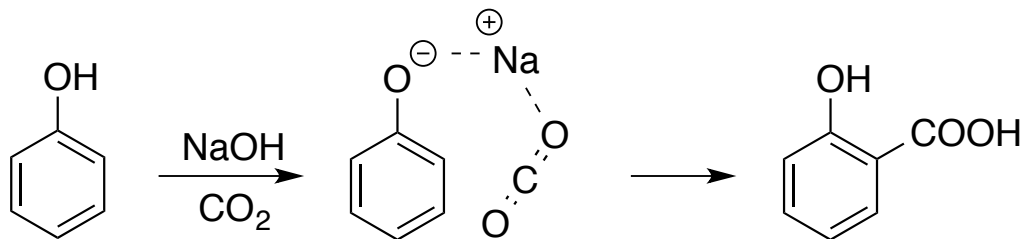
o-Lithiation (Snieckus): Directed Ar-Li formation and reaction with electrophiles.



Directing groups: Amide, Imine, Amine, Ether, Sulfonamide,...

Electrophiles: X_2 , RX , RCHO ,...

Kolbe Schmitt: Reaction of Phenolates with CO_2 (which is a relatively weak electrophile).



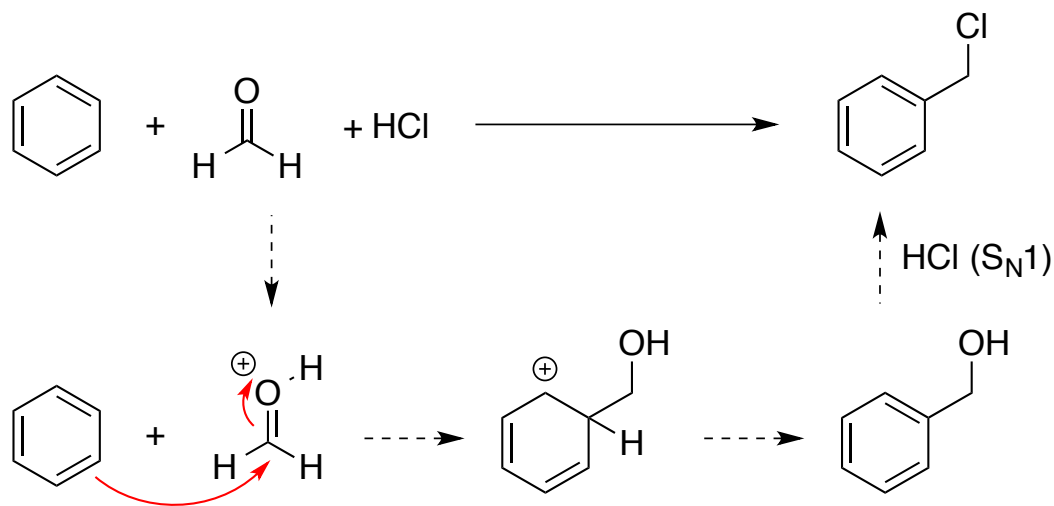
Na^+ is o-directing, larger cations like e.g. Cs^+ lead to para-selectivity

Reactivity of Aromatic Compounds

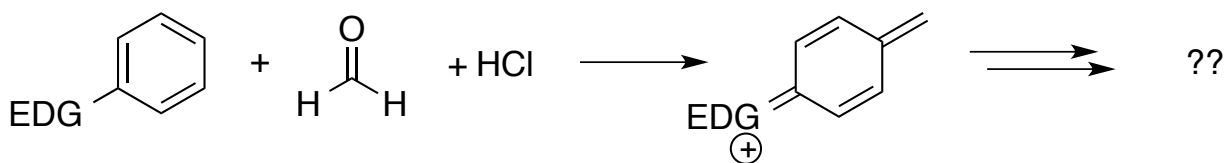
Electrophilic Aromatic Substitution

Formaldehyde as an electrophile:

Chloromethylation:



Activated Aryls:

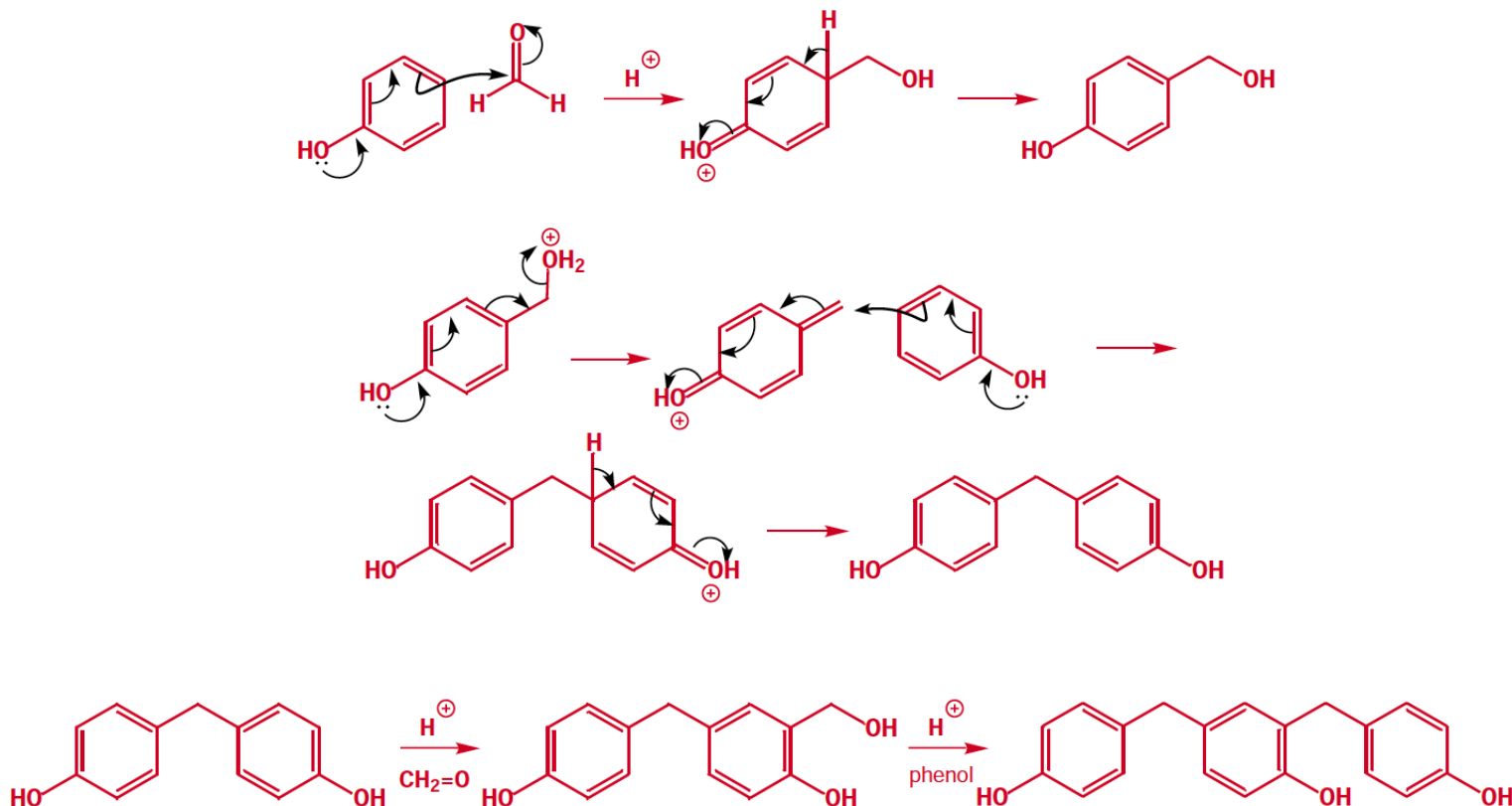


Reactivity of Aromatic Compounds

Electrophilic Aromatic Substitution

Formaldehyde as electrophile: Polymerization of activated aromatics under acidic cond.

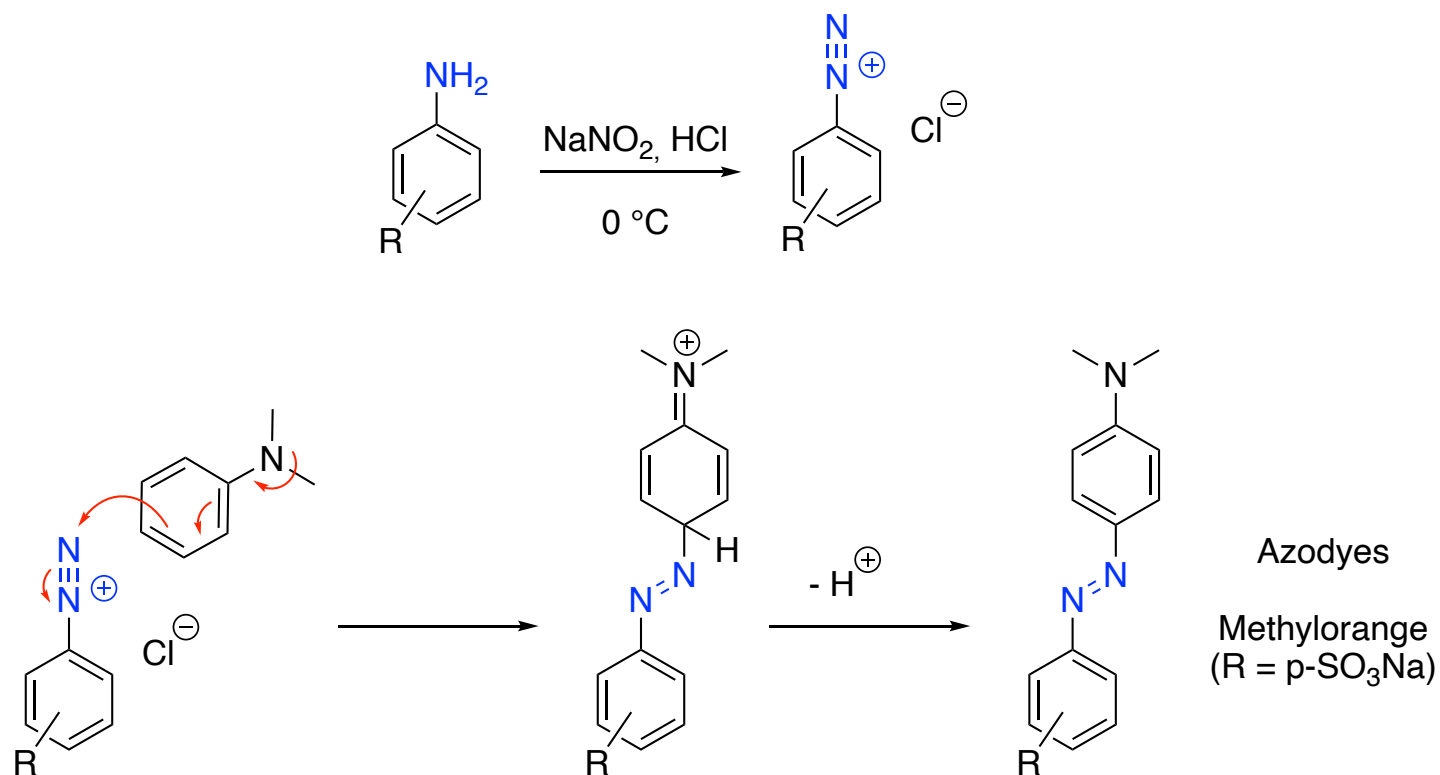
E.g.. Synthesis of **Bakelites** (1905, first fully synthetic organic polymers)!



Reactivity of Aromatic Compounds

Electrophilic Aromatic Substitution

Azo-Coupling: Diazonium salts as electrophiles.



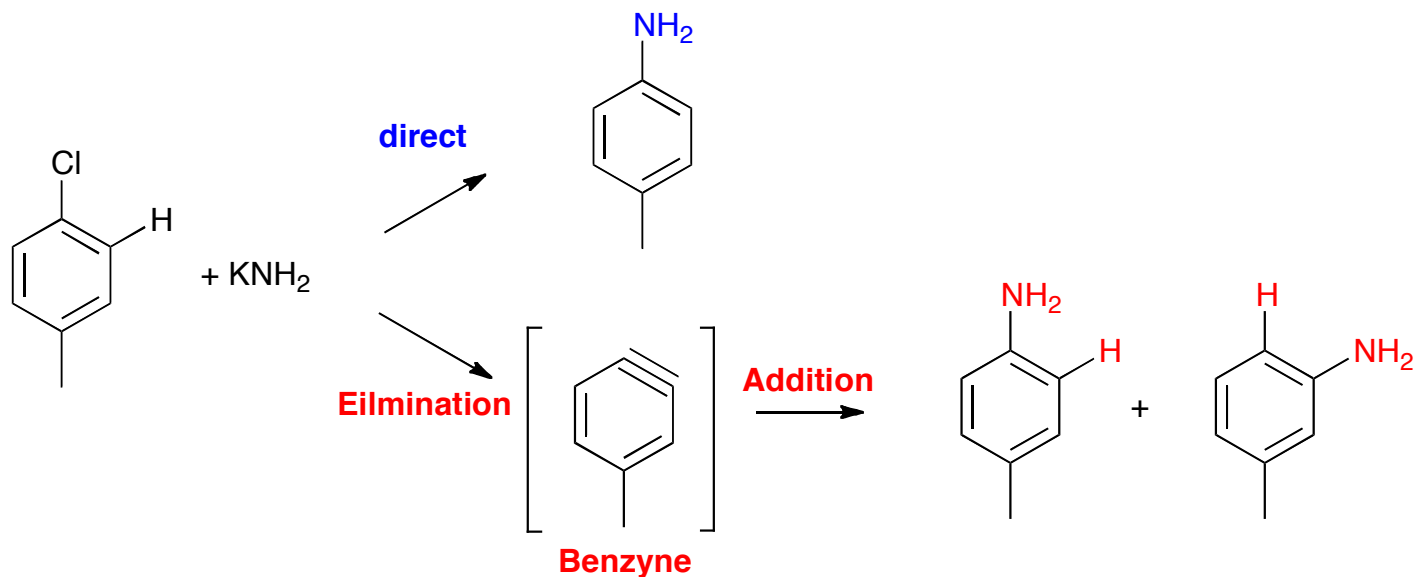
Reactivity of Aromatic Compounds

Nucleophilic Aromatic Substitution

The direct nucleophilic Addition to aromatic systems is rare because of the high electron density of the system and the hindered HOMO-LUMO overlap.

2 common possibilities:

- a) Substitution via formation of an aryne
- b) Addition/elimination on electron-poor aromatics (S_NAr Reaction)

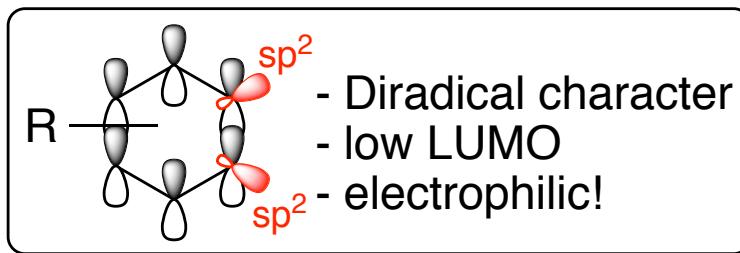
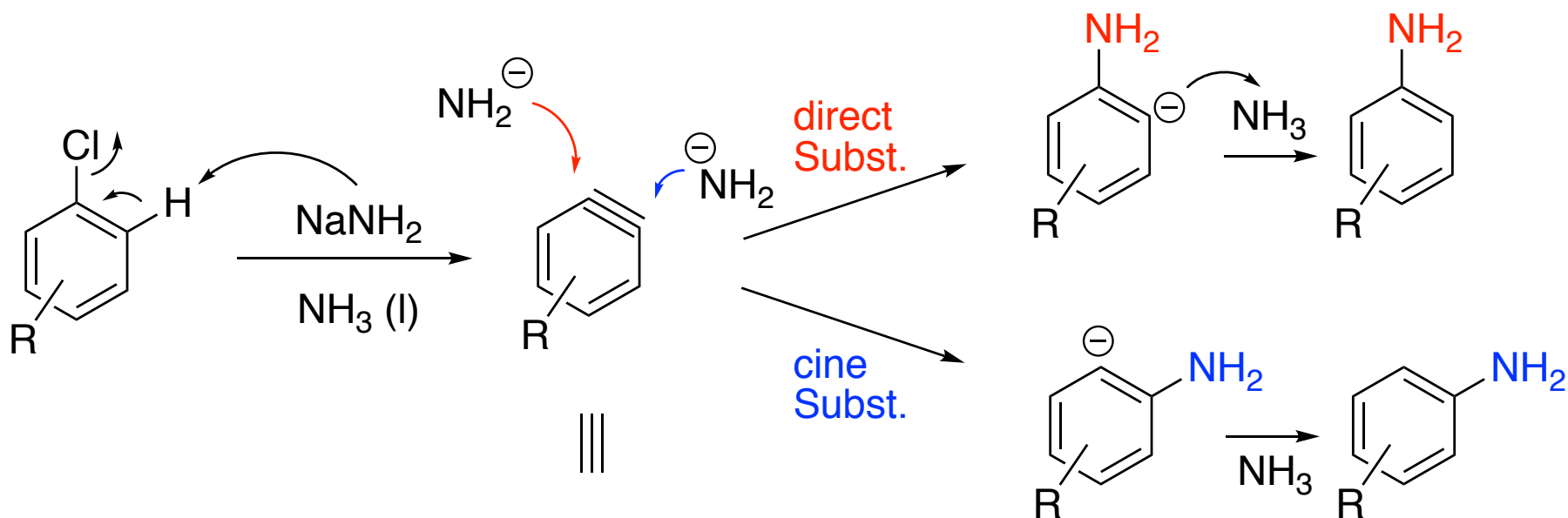


Addition either **direct** (*unlikely*) or via an **Aryne/Benzyne** intermediate (**Elimination – Addition, dominant**) in the presence of a strong base.

Reactivity of Aromatic Compounds

Nucleophilic Aromatic Substitution

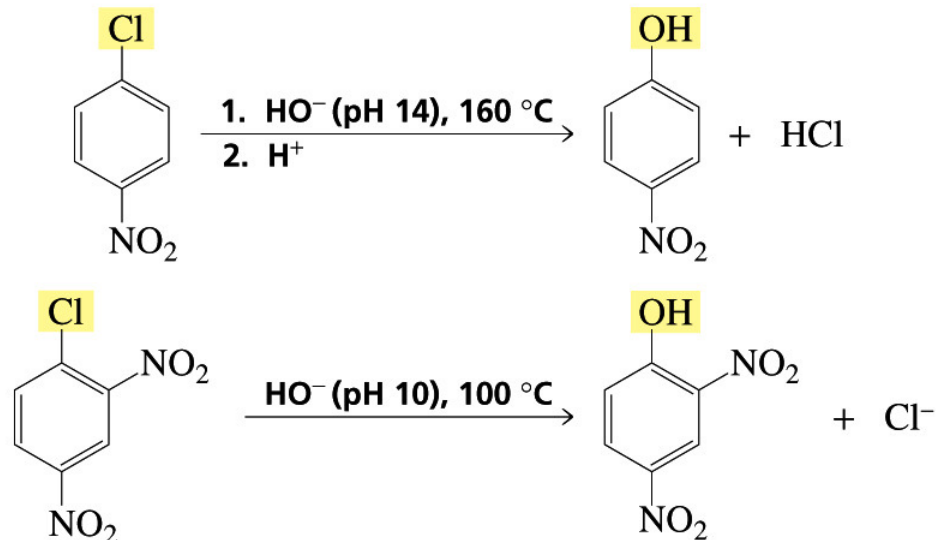
Arynes (i.e. Dehydrobenzene „Benzyne“) are highly reactive intermediates that can be accessed under strongly basic conditions from arylhalides. They have formal triple bond character!



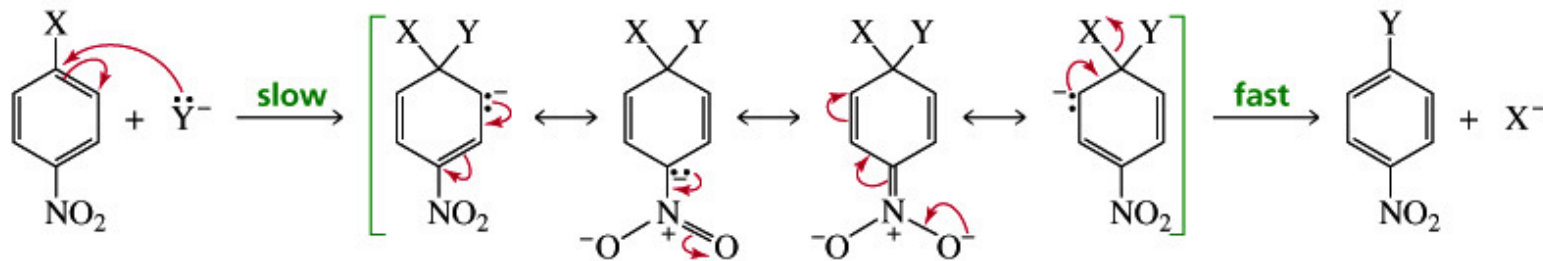
Reactivity of Aromatic Compounds

Nucleophilic Aromatic Substitution

Direct substitution only possible with very **electron-poor** substrates (**S_NAr**)!
(e.g.: Nitro-substituted)



The delocalization of the e⁻-pair of the Nu over the aromatic system aided by the substituents is crucial (M-Effect)! (**conjugated addition – elimination mechanism**)

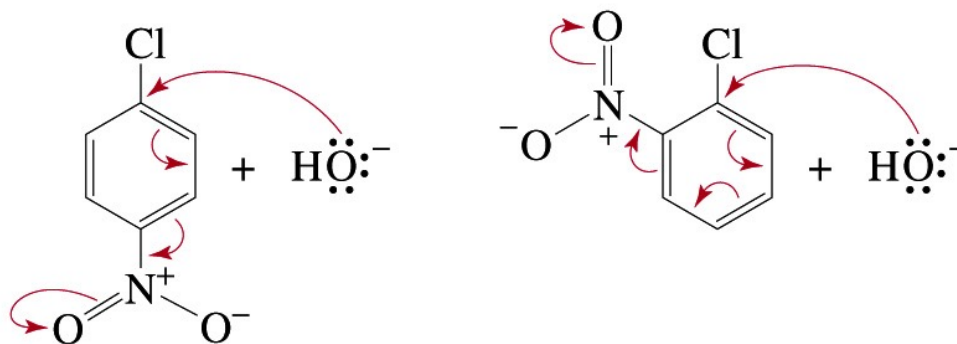


Meisenheimer complexes

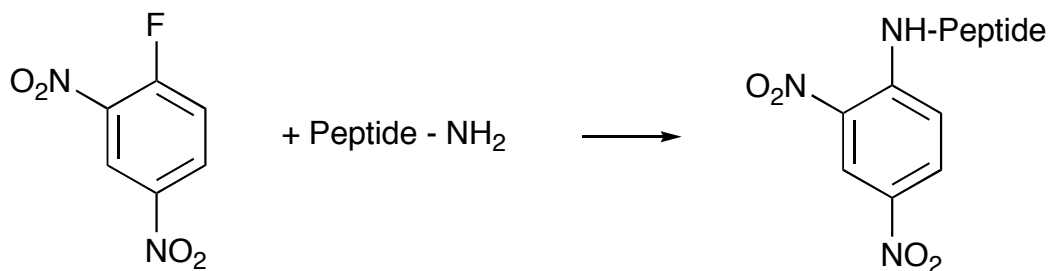
Reactivity of Aromatic Compounds

Nucleophilic Aromatic Substitution

EWG has to be **ortho** or **para** to the leaving group.



Sanger's - Reagent: Important tool in proteine/amino acid analysis (gives a coloured prod.)!

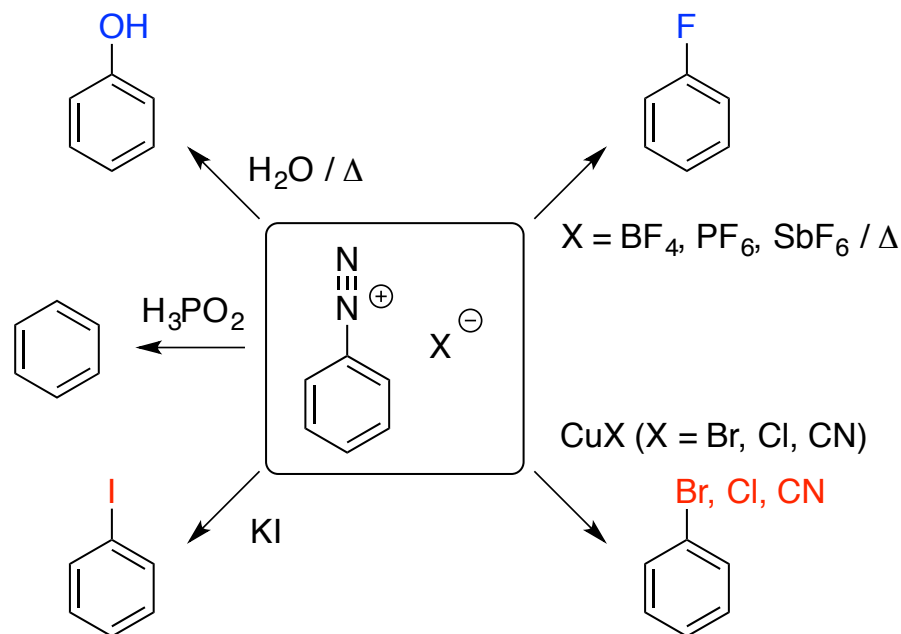


Sanger's reagent

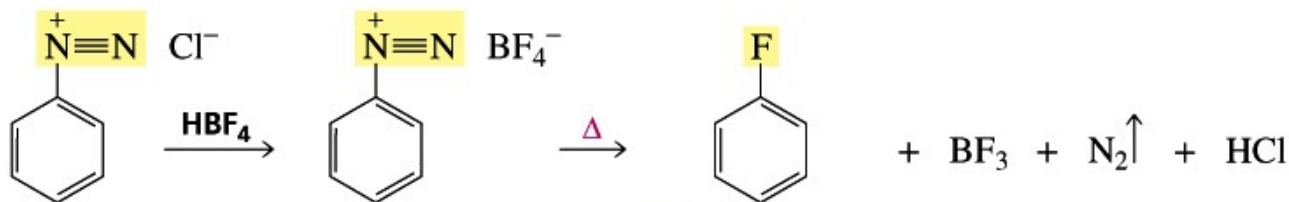
Reactivity of Aromatic Compounds

Nucleophilic Aromatic Substitution

Use of **diazonium** salts:



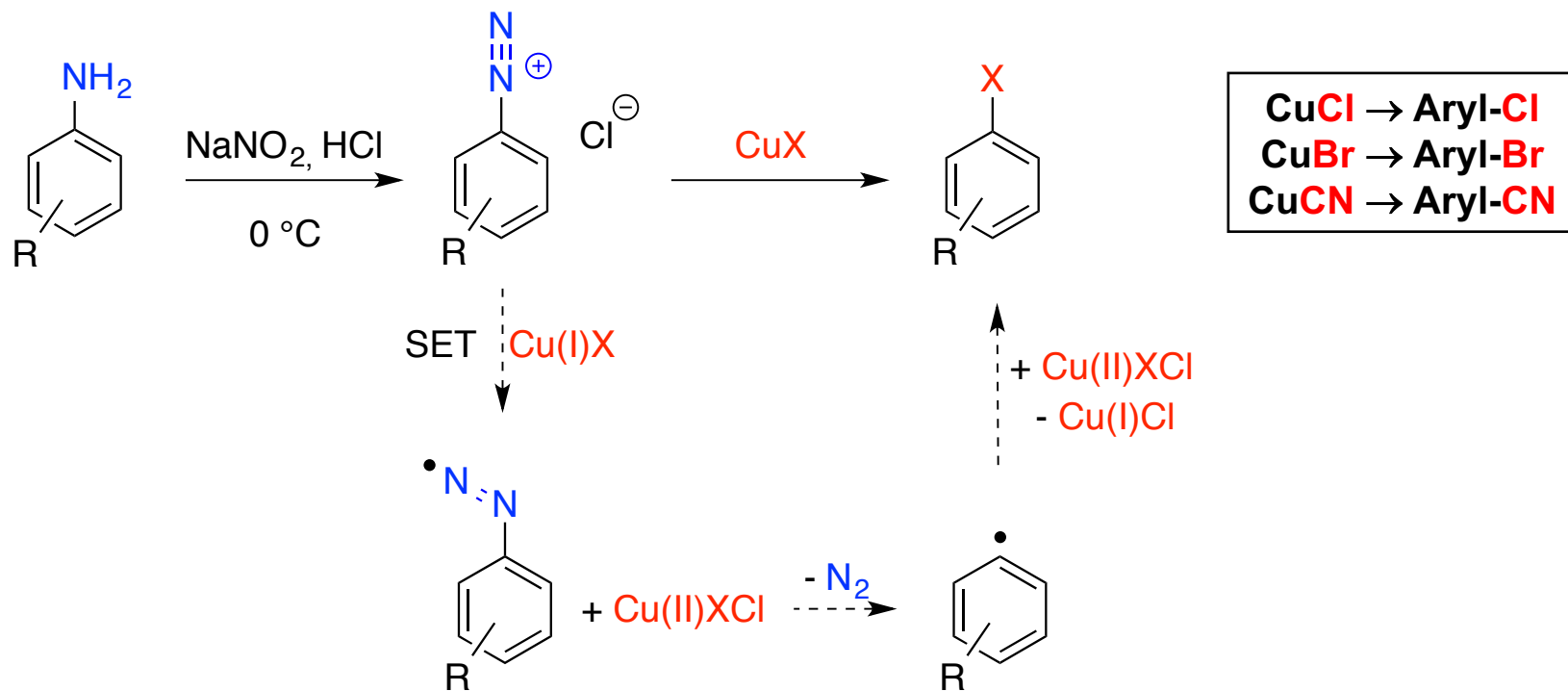
Balz-Schiemann Reaction: Proceeds via a cationic intermediate!



Reactivity of Aromatic Compounds

Nucleophilic Aromatic Substitution

Sandmeyer Reaction: Synthesis of substituted arylhalides from aniline derivatives.
Choice of the **Cu(I)-Salt** crucial (Radicalmech.!)



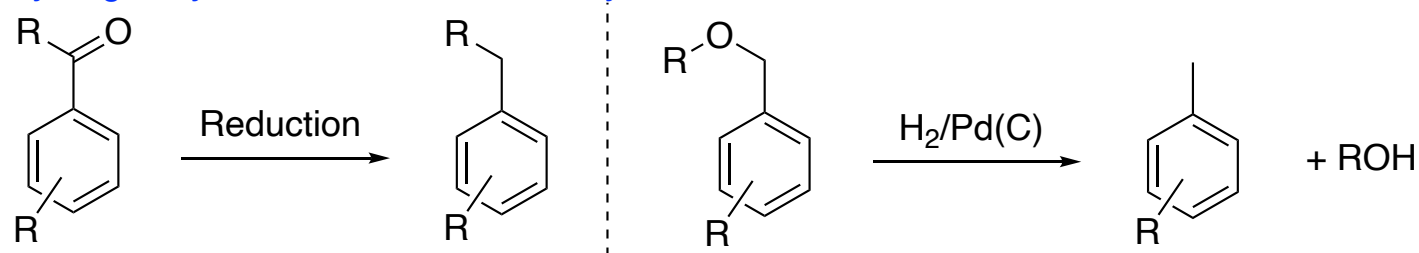
Direct use of KI possible (without Cu-salt!)

Reactivity of Aromatic Compounds

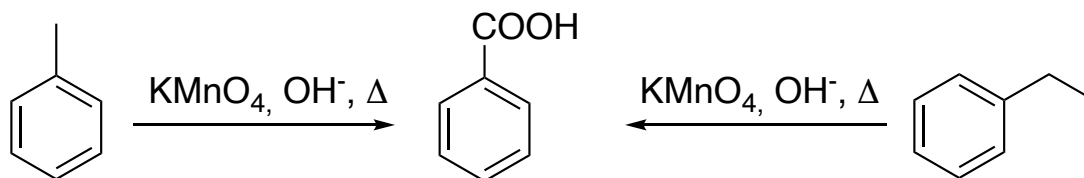
Side Chain Transformations

High reactivity (versatility) of the benzylic position (S_N -Reactions)!

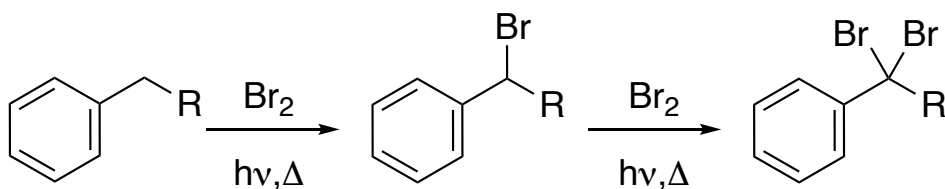
Hydrogenolytic Reduction of the benzylic Pos.



Oxidative Cleavage of the benzylic Pos.



Radical Bromination of the benzylic Pos.



Rule of Thumb for Halogenations:

SSS...Sonnenlicht, Siedehitze, Seitenkette
KKK...Kälte, Katalysator, Kern